A FLAT SEPARATED MORPHISM INJECTIVE ON GENERIC POINTS IS UNIVERSALLY INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof writes X x_S X at one point where the context requires X x_Y X (the author's own source reads 'a generic point eta in X x_S X of an irreducible component of X x_S X'); this is preserved verbatim, with the observation recorded here rather than corrected.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Universal injectivity is equivalent to surjectivity of the diagonal. Morphisms, Lemma morphisms-lemma-universally-injective, invoked at proof lines 6688-6689. Reformulates the goal as surjectivity of Delta.
- Base change of a flat morphism is flat, and generalisations lift along flat morphisms. Morphisms, Lemmas morphisms-lemma-base-change-flat and morphisms-lemma-generalizations-lift-flat, invoked at proof lines 6694-6699. Makes the first projection flat and produces the generic point xi from eta.

Source reference. Original lemma 01U9: sources/stacks-chow/tags.txt, source lines 2402–2402. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03HV: sources/stacks-chow/tags.txt, source lines 4581–4581. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01S4: sources/stacks-chow/tags.txt, source lines 2325–2325. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let X^0 denote the set of generic points of irreducible components of X. If*

(1) *f is flat and separated,*

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(2) *for $\xi \in X^0$ we have $\kappa(f(\xi)) = \kappa(\xi)$, and*
(3) *if $\xi, \xi' \in X^0$, $\xi \neq \xi'$, then $f(\xi) \neq f(\xi')$,*

then f is universally injective.

Proof. We have to show that $\Delta : X \to X \times_Y X$ is surjective, see Morphisms, Lemma 01S4. As $X \to Y$ is separated, the image of Δ is closed. Thus if Δ is not surjective, we can find a generic point $\eta \in X \times_S X$ of an irreducible component of $X \times_S X$ which is not in the image of Δ. The projection $\mathrm{pr}_1 : X \times_Y X \to X$ is flat as a base change of the flat morphism $X \to Y$, see Morphisms, Lemma 01U9. Hence generalizations lift along pr_1, see Morphisms, Lemma 03HV. We conclude that $\xi = \mathrm{pr}_1(\eta) \in X^0$. However, assumptions (2) and (3) guarantee that the scheme $(X \times_Y X)_{f(\xi)}$ has at most one point for every $\xi \in X^0$. In other words, we have $\Delta(\xi) = \eta$ a contradiction. □